

Dynamical crossover at $T^* \simeq 0.55 T_{\text{BKT}}$ in an hBN-encapsulated $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$ weak link, consistent with self-heating-limited switching

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We analyze the zero-field current-driven transport of a single hBN-encapsulated exfoliated $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$ (Bi-2212) weak link measured at 27 temperatures between 2 and 120 K. Resistance–temperature curves at four bias currents (0.316–10 μA) collapse onto a Halperin–Nelson tail giving $T_{\text{BKT}} = 86.7$ K, with a fit-window systematic of ± 0.3 K and a bias spread of ± 0.2 K. Describing every current–voltage branch by self-normalized, threshold-free lineshape descriptors and applying principal-component analysis, we find that all temperatures fall on a one-dimensional manifold whose leading coordinate exhibits a change point at $T^* = 48$ K (bootstrap 68% interval 38–48 K), i.e. $T^*/T_{\text{BKT}} \simeq 0.55$. Three further, independently constructed observables—the supercurrent collapse rate $|dI_c/dT|$, the switching–retrapping hysteresis $1 - I_r/I_c$, and the supercurrent nonreciprocity $|\eta|$ —peak at 45–50 K; a joint bootstrap over repeated sweeps shows that all four peak locations are reproducible to below the 5-K temperature grid, so the four maxima coincide within one grid step. The hysteresis minimum $I_r/I_c = 0.74$ at 50 K is stable in position for voltage criteria spanning 0.3–2 mV, and a polarity-symmetric/antisymmetric decomposition—validated on synthetic data—shows its symmetric (intrinsic) component is recovered without bias while bounding inter-sweep thermal drift at $\lesssim 0.3$ K. A power-balance analysis of the measured retrapping points yields an effective thermal conductance $G \sim 0.01\text{--}0.13$ $\mu\text{W/K}$ which, fed back into the model, reproduces the observed non-monotonic $I_r/I_c(T)$ including the 50-K minimum; a capacitive (RCSJ) origin would instead require an unphysical junction capacitance of ~ 0.04 aF and predicts $I_r/I_c \sim 10^{-3}$. The observations are therefore most naturally described by self-heating-limited (hot-spot) switching dynamics crossing over to overdamped flux flow at T^* . The analysis protocol—concordance of independently constructed, uncertainty-quantified observables extracted by a threshold-free pipeline—is designed for the single-device limit common to air-sensitive two-dimensional superconductors.

I. INTRODUCTION

Atomically thin crystals of the high- T_c cuprate $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$ (Bi-2212) retain superconductivity down to the monolayer limit [1], making them a natural platform for studying two-dimensional (2D) superconductivity, where the resistive transition is governed by Berezinskii–Kosterlitz–Thouless (BKT) vortex physics [2, 3], and, more recently, for nonreciprocal supercurrent phenomena [4, 5]. Because few-unit-cell Bi-2212 degrades within minutes in ambient conditions, such experiments are typically restricted to a small number of devices—often a single one—so the central experimental question becomes: *what can be established rigorously from one device?* Claims that rest on a single extracted number (a critical temperature, a diode efficiency) are vulnerable to threshold choices and measurement artifacts. Here we take a different route: we extract several independently constructed observables from a dense temperature series of current–voltage (I – V) characteristics, quantify the uncertainty of each by resampling, quantify their residual statistical correlations, and ask whether they agree.

We find that they do. Four observables—an unsupervised, threshold-free I – V lineshape coordinate, the supercurrent collapse rate, the switching–retrapping hysteresis, and the supercurrent nonreciprocity—identify the same crossover band, $T^* = 45\text{--}50$ K $\simeq 0.55 T_{\text{BKT}}$,

far below the resistive transition at $T_{\text{BKT}} = 86.7$ K. The phenomenology on both sides of T^* , in particular the non-monotonic temperature dependence of the retrapping-to-switching ratio I_r/I_c , is quantitatively consistent with hysteresis governed by self-heating (hot-spot formation) rather than junction capacitance [6–9]: a power-balance model calibrated on the measured retrapping points reproduces $I_r/I_c(T)$ including its 50-K minimum (Appendix C), while a capacitive origin fails by more than three orders of magnitude. We phrase the interpretation as “consistent with” throughout: a direct thermal measurement (pulsed I – V or on-chip thermometry) remains the decisive test.

II. DEVICE AND MEASUREMENTS

The device is a mechanically exfoliated Bi-2212 flake transferred onto a SiO_2 (285 nm)/Si substrate, contacted by four Au electrodes and capped by an hBN flake that protects the crystal from ambient degradation. Two families of zero-field measurements are analyzed: (i) four-terminal I – V sweeps at 27 temperatures between 2 and 120 K, with the current ramped between ± 0.1 mA in 1- μA steps under a ± 0.4 V compliance, in both ramp directions (from $+I_{\text{max}}$ to $-I_{\text{max}}$ and back), with two to five repetitions per temperature (131 sweeps in total); and (ii) resistance–temperature $R(T)$ curves through the

transition at fixed bias currents of 0.316, 1, 3.16, and 10 μA . Control sweeps at 10 K taken ~ 30 times slower (2 s per point) reproduce the standard-rate switching current to within 3%, bounding sweep-rate effects at that level; repeated sweeps at fixed temperature reproduce I_c with a median scan-to-scan spread of 0.05%, demonstrating the stability of the hBN-protected device over the full campaign.

All analysis is produced by a deterministic, configuration-driven pipeline (Appendix A). Raw sweeps are split into rising- and falling- $|I|$ legs; because the ramp direction and current sign fix whether $|I|$ is increasing or decreasing, each measured point belongs unambiguously to one of four branches: switching ($|I|$ rising) or retrapping ($|I|$ falling), at positive or negative polarity. With this protocol the rising leg at positive polarity and the rising leg at negative polarity occur in *different* sweep files; all cross-file comparisons are therefore explicitly drift-bounded below. Quality control adds boolean flags (voltage rail, noise floor $\sim 1.5 \times 10^{-4}$ V) but never deletes or modifies data. Uncertainties on derived quantities are nonparametric bootstrap intervals over repeated sweeps, and threshold-dependent quantities carry an explicit sensitivity band.

III. RESULTS

A. Transition temperature and its systematics

Figure 1(c) shows the resistive transition at four bias currents spanning a factor of 30. The rising tail is described by the Halperin–Nelson (HN) form [3]

$$R(T) = R_0 \exp\left[-b(T - T_{\text{BKT}})^{-1/2}\right], \quad (1)$$

with $b = 2.5\text{--}3.0$. Statistical uncertainties on T_{BKT} are below 0.03 K; the relevant uncertainties are systematic, and we quantify them in Appendix B: varying the fit window over [85.5–87, 93–97] K moves T_{BKT} within 86.4–87.0 K, the spread across bias currents is 0.2 K, the fitted b and T_{BKT} are strongly anticorrelated ($r = -0.91$), and the residuals show a coherent $\pm(0.02\text{--}0.04)\Omega$ structure common to all four currents, indicating a small departure from the pure HN form as expected for an inhomogeneity-broadened transition [13]. We therefore quote $T_{\text{BKT}} = 86.7 \pm 0.3$ K (systematic). The complementary I – V criterion $V \propto I^{\alpha(T)}$, $\alpha(T_{\text{BKT}}) = 3$, is not robust in this device: the accessible dissipative window is truncated from below by the noise floor and from above by the switching instability, and the apparent $\alpha = 3$ crossings move or vanish as the fit window is varied. We rely on the HN tail, which is threshold-free and reproducible across bias.

B. Threshold-free lineshape analysis and the crossover T^*

Figure 1(a) shows why any fixed voltage criterion is dangerous here: at 5 K the switching branch jumps discontinuously to hundreds of mV; at 35–55 K a gradual low-voltage foot precedes the jump; near T_{BKT} the response is smooth. A fixed millivolt-scale threshold therefore samples *different physical features* at different temperatures.

To characterize the evolution objectively we describe each rising- $|I|$ branch by six *self-normalized* descriptors—defined precisely in Appendix A—none of which contains an absolute voltage scale. Principal-component analysis of the standardized descriptors [Fig. 2(a)] places the 27 temperatures on a single ordered arc (PC1 and PC2 carry 43% and 36% of the variance), so the I – V lineshape evolution is governed by essentially one latent coordinate. We deliberately do *not* cluster these points: on a one-dimensional manifold, hard clustering would impose sharp boundaries on what the data show to be a continuous evolution.

The leading coordinate PC1 rises steeply between ~ 35 and ~ 55 K [Fig. 2(b)]. A least-squares change-point analysis of PC1(T), bootstrapped over repeated sweeps (300 resamples), locates the crossover at $T^* = 48$ K with a 68% interval of 38–48 K; the bootstrap distribution [inset] is unimodal and contains no mass near T_{BKT} .

C. Switching, retrapping, and intrinsic hysteresis

Figure 3(a) shows the switching current I_c and retrapping current I_r (voltage criterion $V_{\text{th}} = 1$ mV, averaged over polarities and repeated sweeps). From $I_c = 64.0 \mu\text{A}$ at 2 K, the supercurrent collapses steeply between 35 and 55 K, passes through a shoulder at 55–75 K near $10 \mu\text{A}$, and vanishes at T_{BKT} . The ratio I_r/I_c [Fig. 3(b)] is flat at 0.93 below 30 K, drops to a minimum of 0.74 at 50 K, and recovers to $\simeq 1$ above 60 K. The position of this minimum is not an artifact of the voltage criterion: for V_{th} between 0.3 and 2 mV—almost a decade—the minimum stays at 50–55 K, i.e. within one step of the temperature grid [Appendix B, Fig. 5(a)]. (At $V_{\text{th}} = 5$ mV the criterion enters the flux-flow foot and no longer measures the switching instability; this pathology is documented in Appendix B.)

Because I_c and I_r at a given polarity are extracted from sweeps taken in opposite ramp directions (Sec. II), an apparent hysteresis could be produced by a temperature offset δT between the two sweep files, of magnitude $\Delta I \approx |dI_c/dT| \delta T$. We separate the two by symmetry: a switching–retrapping hysteresis is *even* under current-polarity reversal, whereas a drift artifact is *odd*. Writing $h_{\pm} = I_{\pm}^{\pm} - I_r^{\pm}$,

$$h_{\text{sym}} = \frac{1}{2}(h_{+} + h_{-}), \quad h_{\text{anti}} = \frac{1}{2}(h_{+} - h_{-}). \quad (2)$$

As shown in Fig. 4(a), h_{sym} carries 92% of the total and

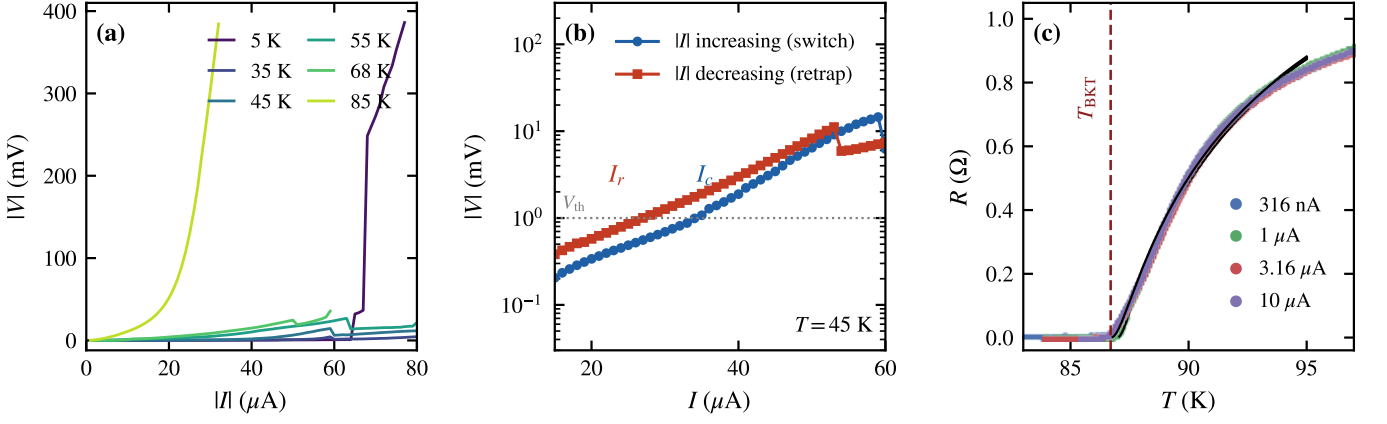


FIG. 1. Transport phenomenology of the hBN-capped Bi-2212 weak link. (a) Rising- $|I|$ (switching) branches of the I - V characteristic at representative temperatures. Three behaviors are visible: an abrupt switch to the dissipative state at low T (5 K), a gradual low-voltage “flux-flow foot” at intermediate T (35–55 K), and a smooth, nearly ohmic response near the transition (85 K). (b) Switching (blue) versus retrapping (red) branches at 45 K on a logarithmic voltage scale; the two branches cross the $V_{\text{th}} = 1$ mV criterion (dotted) at different currents, I_c and I_r , revealing pronounced hysteresis. (c) $R(T)$ at four bias currents with Halperin–Nelson fits [Eq. (1), black]; the fits give $T_{\text{BKT}} = 86.7$ K with a fit-window systematic of ± 0.3 K (dashed line; Appendix B).

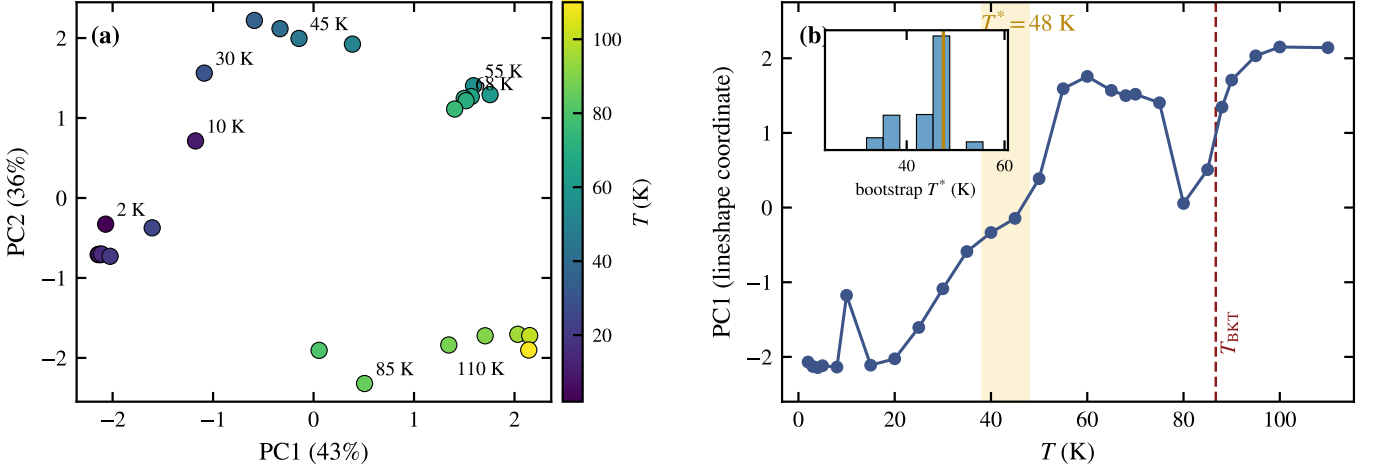


FIG. 2. Unsupervised lineshape analysis. (a) Principal-component scores of the six self-normalized I - V descriptors (Appendix A); each point is one temperature. (b) The leading coordinate PC1 versus temperature. The shaded band marks the bootstrap change point $T^* = 48$ K (68% interval 38–48 K); the dashed line marks T_{BKT} . Inset: bootstrap distribution of the change point.

peaks at $7.0 \mu\text{A}$ at 45 K, while h_{anti} is small and uncorrelated with $|dI_c/dT|$ ($r = -0.03$); interpreted as drift it corresponds to $\delta T \lesssim 0.3$ K. We validated the decomposition on synthetic data [Appendix B, Fig. 5(c)]: with a known injected $\delta T = 0.3$ K the estimator recovers 0.30 K and the intrinsic component h_{sym} is unbiased to $< 0.2\%$; when an adversarial intrinsic *polarity asymmetry* of the hysteresis is added, h_{sym} remains unbiased but h_{anti} absorbs the asymmetry. Accordingly, we claim only that (i) the intrinsic hysteresis is measured without bias, and (ii) h_{anti} bounds the *combined* magnitude of thermal drift and any intrinsic polarity asymmetry at the equivalent of 0.3 K—not that drift is “fully excluded.”

The supercurrent nonreciprocity $\eta = (I_c^+ - |I_c^-|)/(I_c^+ +$

$|I_c^-|)$ on the switching branch also compares quantities from different sweep files, and inherits the same drift bound: at 50 K, $|dI_c/dT| \delta T / (2I_c) \lesssim 1.5\%$. The measured value rises from $|\eta| < 0.5\%$ below 40 K to 6.2% (bootstrap 95% interval 4.2–8.2%) at 50 K, so at least $\sim 3/4$ of the peak value is not attributable to drift. Without magnetic-field-reversal data we make no claim about its microscopic origin—in particular no superconducting-diode claim [4, 5]; we report it as a bounded observable that peaks in the T^* band.

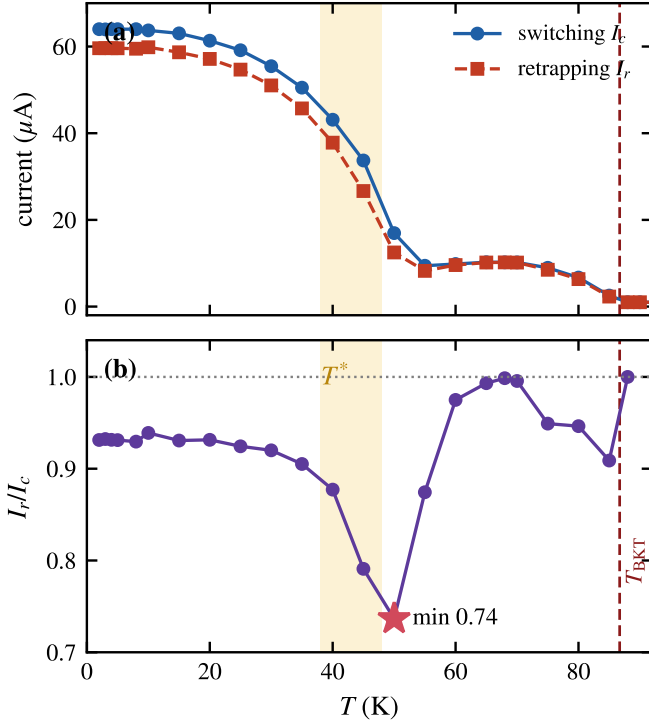


FIG. 3. (a) Switching current I_c and retrapping current I_r versus temperature ($V_{\text{th}} = 1$ mV; polarity- and sweep-averaged). (b) The ratio I_r/I_c : hysteresis is maximal ($I_r/I_c = 0.74$) at 50 K, in the T^* band (shading), and vanishes above 60 K. The dashed line marks T_{BKT} .

D. Concordance at T^*

Figure 4(b) overlays, on a common normalized scale, the four observables: $|dI_c/dT|$, $1 - I_r/I_c$, $|\eta|$, and the lineshape change rate $|d\text{PC1}/dT|$. A *joint* bootstrap—resampling the repeated sweeps once per iteration and re-deriving all four curves and their extrema simultaneously (400 resamples)—gives peak locations of 45 K ($|dI_c/dT|$) and 50 K (the other three) in every single resample: the statistical uncertainty of each peak position is below the 5-K spacing of the temperature grid, and the appropriate uncertainty to quote is the grid half-spacing, ± 2.5 K. All four maxima therefore coincide within one grid step.

These observables are *independently constructed*, but not statistically independent, since several share the underlying switching currents. We quantify this in Appendix B: the curve-level Pearson correlations over 2–88 K are $r = 0.89$ for the pair ($|dI_c/dT|$, $1 - I_r/I_c$), which share I_c , but only $r = 0.26$ – 0.48 for all pairs involving $|\eta|$ or $|d\text{PC1}/dT|$. In particular the lineshape coordinate contains no absolute current scale at all, so the agreement of its change point with the I_c -based extrema is a genuine cross-check rather than a restatement. (The lineshape rate shows a secondary peak at T_{BKT} , where the I – V curves reorganize again at the resistive transition, as expected.)

IV. DISCUSSION

A. A thermal origin of the crossover

The non-monotonic hysteresis is the most discriminating clue to the physics at T^* . In the resistively-and-capacitively-shunted-junction (RCSJ) picture [10, 11], hysteresis requires an underdamped phase and decreases monotonically with temperature. Quantitatively, RCSJ fails here on three counts (Appendix C): with the measured dissipative-branch resistance $R_d \simeq 15$ k Ω and any realistic geometric capacitance (0.1–1 pF), the predicted ratio is $I_r/I_c \sim (2\text{--}6) \times 10^{-4}$, three orders of magnitude below the observed 0.74–0.93; reproducing the observed ratio would require $C \simeq 0.04$ aF; and the predicted temperature dependence is monotonic. Thermally activated escape [12] is likewise negligible: $E_J/k_B = \hbar I_c/2ek_B \approx 1.5 \times 10^3$ K at 2 K and $\approx 4 \times 10^2$ K at 50 K, so $k_B T/E_J \lesssim 0.12$ throughout.

By contrast, a self-heating description [6–9] is quantitatively consistent with the data. At each temperature the measured retrapping point (I_r, V_r) defines, through the power balance $P = I_r V_r = G(T_j - T_{\text{bath}})$ with T_j fixed by $I_r = I_c(T_j)$, an effective thermal conductance G between the dissipative region and the bath. The implied G falls in the range 0.08–0.13 $\mu\text{W/K}$ where the device retraps from the fully resistive branch ($T \leq 15$ K, hot-spot excursions $T_j - T_{\text{bath}} \approx 15$ –20 K) and ~ 0.01 $\mu\text{W/K}$ where it retraps from the mV-scale flux-flow state (20–50 K)—both plausible magnitudes for a micrometer-scale flake anchored through ~ 300 nm of SiO_2 , though the factor-of-ten split shows that a single- G hot-spot model is an oversimplification. Feeding a smooth interpolation of the calibrated $G(T)$ back into the model *predicts* $I_r/I_c(T)$ in agreement with the measurement, including the 50-K minimum [Appendix C, Fig. 6(b)]. In this picture the regimes separated by T^* acquire a simple meaning: below T^* , switching drives the bridge into a self-sustaining hot spot (voltages jump to the 100-mV scale); around T^* the collapsing supercurrent only marginally sustains the hot spot—here hysteresis and nonreciprocity are maximal and the flux-flow foot emerges; above T^* , dissipation sets in gradually through vortex flow [14] and the dynamics are reversible up to T_{BKT} . We emphasize that this identification is *consistency*, not proof: pulsed I – V protocols or on-chip thermometry would provide the direct thermal evidence.

B. Role of the hBN encapsulation

Encapsulation quality is a live issue for Bi-2212 devices, whose surface degrades by oxygen out-diffusion and water adsorption. Three properties of the present hBN-capped device bear directly on this: (i) $T_{\text{BKT}} = 86.7 \pm 0.3$ K with a sharp HN tail, compared with zero-resistance temperatures of 75.5–86.4 K reported for uncapped flakes of comparable and larger thickness on

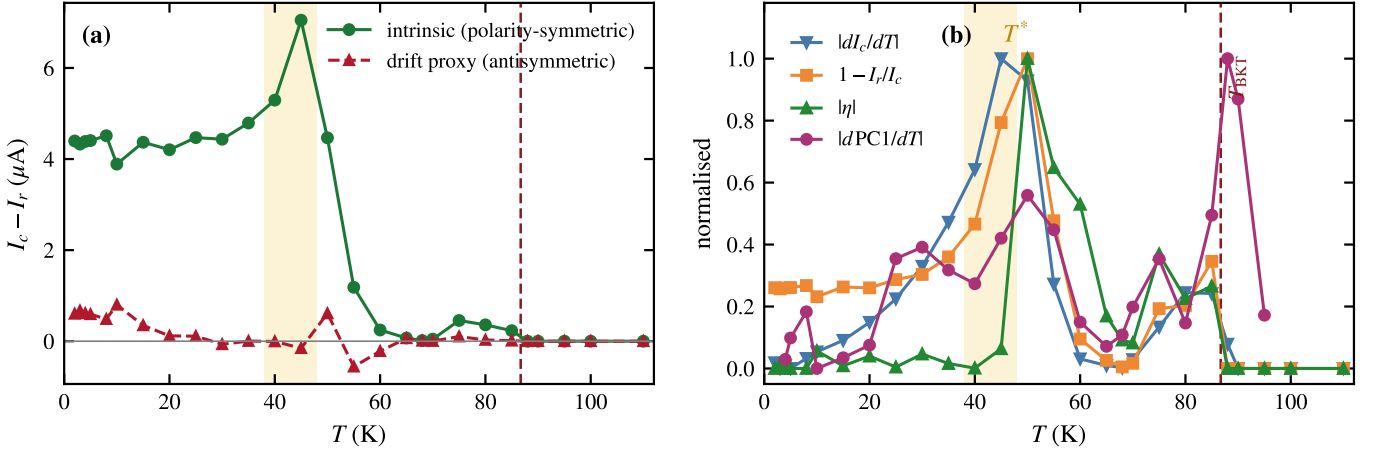


FIG. 4. (a) Polarity-symmetric (intrinsic) and antisymmetric (drift-proxy) components of the switching–retrapping hysteresis $I_c - I_r$ [Eq. (2)]. The symmetric component carries 92% of the signal and peaks at $7.0 \mu\text{A}$ at 45 K ; the antisymmetric component corresponds to $\delta T \lesssim 0.3 \text{ K}$. (b) Concordance of the four observables (each normalized to $[0, 1]$): $|dI_c/dT|$, $1 - I_r/I_c$, $|\eta|$, $|dPC1/dT|$. All four peak in the T^* band (shading); each peak position is reproducible to below the 5-K grid spacing under a joint bootstrap (Appendix B).

plasma-treated SiO_2 [5]; (ii) a median scan-to-scan reproducibility of I_c of 0.05% across 131 sweeps spanning the full campaign, with no monotonic aging trend; and (iii) an inter-sweep temperature-equivalent stability bound of 0.3 K from the antisymmetric hysteresis channel. A single device cannot separate encapsulation from flake-to-flake variation, but these numbers set a useful stability benchmark for hBN capping against which other passivation schemes (uncapped, polymer-capped, or oxide-capped) can be compared on equal statistical footing using the released pipeline.

V. CONCLUSION

From a dense zero-field transport dataset on one hBN-encapsulated Bi-2212 weak link we have shown that (i) the resistive transition sits at $T_{\text{BKT}} = 86.7 \pm 0.3 \text{ K}$ (systematic), reproducible across a 30-fold range of bias current, with fully disclosed fit-window systematics; (ii) a threshold-free, unsupervised analysis of the I – V line-shape reveals a one-dimensional evolution with a change point at $T^* = 48 \text{ K} \simeq 0.55 T_{\text{BKT}}$; (iii) the switching–retrapping hysteresis is intrinsic—its symmetric component is bias-free under a validated decomposition that also bounds drift at 0.3 K—and maximal at T^* ($I_r/I_c = 0.74$, criterion-stable from 0.3 to 2 mV), where the bounded nonreciprocity (6.2%, of which $\lesssim 1.5\%$ could be instrumental) also peaks; and (iv) a power-balance model calibrated on the measured retrapping points reproduces the non-monotonic $I_r/I_c(T)$, while capacitive and fluctuation mechanisms fail quantitatively by orders of magnitude. Beyond the specific device, the methodology—concordance of independently constructed, resampled observables with explicit sensitivity and confound tests, in place of any single thresholded number—offers a tem-

plate for defensible transport claims in the single-device limit.

DATA AND CODE AVAILABILITY

The complete raw data, processed data, configuration files, analysis pipeline, and figure-generation code are available at <https://github.com/LaipoHsun/BSCCO-superconducting-2D-material-analysis->. Every figure in this paper is regenerated from the raw measurement files by the scripts in `pipeline/`.

ACKNOWLEDGMENTS

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Appendix A: Analysis definitions

Critical currents. For each branch (switching/retrapping $\times$ polarity) of each sweep, I_c (or I_r) is the current at which $|V|$ first crosses $V_{\text{th}} = 1 \text{ mV}$, obtained by linear interpolation between the bracketing samples on the $|I|$ -sorted branch. Aggregates are means over polarities and repeated sweeps; intervals are percentile bootstrap over sweeps (2000 resamples unless stated).

Lineshape descriptors. Each rising- $|I|$ branch with maximum voltage $V_{\text{max}} > 10\sigma_V$ (noise floor $\sigma_V \simeq 1.5 \times 10^{-4} \text{ V}$) is reduced to: (1) I_{50} , the current at which $|V|$ first reaches $0.5 V_{\text{max}}$ (interpolated); (2) $V_{\text{max}}/V_{\text{comp}}$ with $V_{\text{comp}} = 0.4 \text{ V}$ the compliance; (3) the switching width $(I_{90} - I_{10})/I_{50}$, where I_f is the current at

$f V_{\max}$; (4) the dissipationless fraction of samples with $|V| < 5\sigma_V$; (5) the foot fraction of samples with $1 \text{ mV} < |V| < 0.1 V_{\max}$; and (6) the slope dV/dI over the top quartile of the branch in $|I|$. Only ratios of each branch's own scales enter (1), (3)–(5); (2) and (6) are reported for completeness and carry low PCA weight.

PCA and change point. Descriptors are averaged over sweeps at each temperature, standardized to zero mean and unit variance across temperatures, and decomposed by singular-value decomposition; PC1 is sign-oriented to increase with T . The change point of PC1(T) minimizes the total within-segment sum of squares over all binary partitions of the T -ordered sequence; its uncertainty is the 16–84% interval over 300 bootstrap resamples of the per-sweep descriptor sets.

Derivatives. dI_c/dT and $d\text{PC1}/dT$ use second-order central differences on the non-uniform temperature grid (`numpy.gradient`); PC1 is first smoothed with a 3-point boxcar, and the two points at each end of the grid, where the boxcar and difference stencils are one-sided, are excluded from peak searches. Peak searches are restricted to $T \leq 80 \text{ K}$, below the reorganization associated with T_{BKT} itself.

Appendix B: Robustness and systematics

Threshold sensitivity (Fig. 5(a)). The I_r/I_c minimum sits at 50 K for $V_{\text{th}} = 1\text{--}2 \text{ mV}$ and at 55 K for $0.3\text{--}0.5 \text{ mV}$ —one grid step—with depth 0.61–0.78. At 5 mV the criterion lands inside the flux-flow foot [cf. Fig. 1(a)] and the apparent minimum moves to 35 K; this is the documented pathology of large thresholds, not a property of the switching instability.

Halperin–Nelson systematics (Fig. 5(b)). Across a grid of fit windows (lower edge 85.5–87 K, upper edge 93–97 K) and the four bias currents, T_{BKT} spans 86.4–87.0 K. For the baseline window (86–95 K, $1 \mu\text{A}$): $R_0 = 2.10 \Omega$, $b = 2.50$, $T_{\text{BKT}} = 86.92 \text{ K}$, with parameter correlations $r(b, T_{\text{BKT}}) = -0.91$, $r(R_0, b) = +0.99$; residuals show a coherent $\pm(0.02\text{--}0.04) \Omega$ structure shared by all four currents. We quote the window spread, not the (much smaller) statistical errors, as the uncertainty.

Synthetic validation of the drift decomposition (Fig. 5(c)). Synthetic datasets use the measured smoothed $I_c(T)$, the measured intrinsic hysteresis, per-scan noise of $0.3 \mu\text{A}$, and an injected inter-file offset $\delta T = 0.3 \text{ K}$. Over 200 realizations the estimator $\hat{\delta T} = \text{median}[-h_{\text{anti}}/(dI_c/dT)]$ recovers 0.30 K , and h_{sym} is unbiased to $< 0.2\%$. Adding an intrinsic polarity asymmetry of 20% of the hysteresis leaves h_{sym} unbiased

($< 0.1\%$) but shifts $\hat{\delta T}$ to 0.99 K : the antisymmetric channel cannot distinguish drift from intrinsic polarity asymmetry, which is why we quote it only as a combined bound.

Observable correlations. Curve-level Pearson correlations over 2–88 K: $r(|dI_c/dT|, 1 - I_r/I_c) = 0.89$; $r(|dI_c/dT|, |\eta|) = 0.35$; $r(|dI_c/dT|, |d\text{PC1}/dT|) = 0.40$; $r(1 - I_r/I_c, |\eta|) = 0.48$; $r(1 - I_r/I_c, |d\text{PC1}/dT|) = 0.26$; $r(|\eta|, |d\text{PC1}/dT|) = 0.27$. In the joint bootstrap the four peak locations are identical in all 400 resamples (45, 50, 50, 50 K), so their statistical covariance vanishes at the grid resolution and each position carries the grid-limited uncertainty $\pm 2.5 \text{ K}$.

Appendix C: Thermal power-balance model and RCSJ comparison

Implied thermal conductance. On the falling branch, the last dissipative samples before retrapping give the retrapping power $P = |V||I|$ (averaged over the three samples above I_r). Setting $P = G(T_j - T_{\text{bath}})$ with the hot-spot temperature T_j fixed by $I_r = I_c(T_j)$ —inverting the measured, monotone part of $I_c(T)$ ($T \leq 55 \text{ K}$, PCHIP interpolation—yields G at each bath temperature [Fig. 6(a)]. Where the device retraps from the fully resistive branch ($T \leq 15 \text{ K}$, $V_r = 15\text{--}37 \text{ mV}$) we find $G = 0.08\text{--}0.13 \mu\text{W/K}$ with hot-spot excursions of 10–20 K; where it retraps from the mV-scale flux-flow state (20–50 K) the implied $G \approx 0.01 \mu\text{W/K}$ is a lower bound, since V_r there is limited by the 1-mV criterion.

Forward prediction. A smooth power-law interpolation of $G(T)$ is fed back into the retrapping condition evaluated on the measured falling branches: descending in $|I|$, the predicted I_r is the first current at which $I_c(T_{\text{bath}} + |V||I|/G) \geq |I|$. The result [Fig. 6(b)] reproduces the measured $I_r/I_c(T)$ including the non-monotonic minimum at 50 K with no further adjustment.

RCSJ comparison. The measured dissipative-branch slope at 2–5 K is $R_d = 15.4 \text{ k}\Omega$. In RCSJ the quality factor is $Q = \sqrt{2eI_c R_d^2 C / \hbar}$ and, for $Q \gg 1$, $I_r/I_c \simeq 4/\pi Q$. With $I_c = 64 \mu\text{A}$ and a geometric $C = 0.1 \text{ pF}$ (1 pF), $Q \simeq 2.2 \times 10^3$ (6.8×10^3), predicting $I_r/I_c \simeq 6 \times 10^{-4}$ (2×10^{-4})—three orders of magnitude below the measured 0.93. Reproducing 0.93 would require $C \simeq 0.04 \text{ aF}$, orders of magnitude below any physical junction capacitance. Capacitive hysteresis is therefore excluded quantitatively, and its predicted monotonic temperature dependence is additionally inconsistent with the observed minimum.

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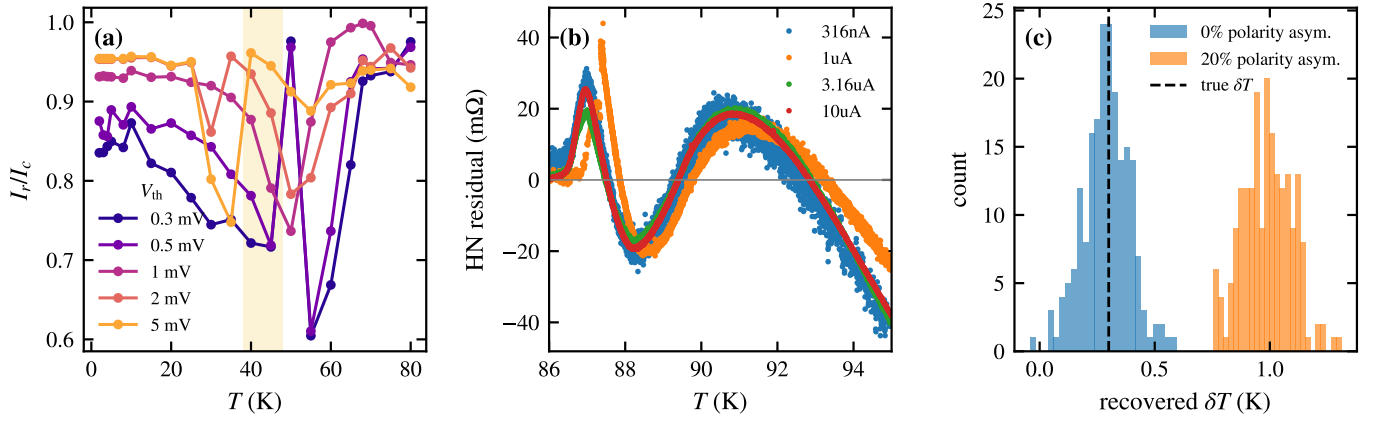


FIG. 5. Robustness tests. (a) $I_r/I_c(T)$ for voltage criteria 0.3–5 mV; the minimum stays at 50–55 K for 0.3–2 mV, while 5 mV enters the flux-flow foot (see text). (b) Halperin–Nelson fit residuals for the four bias currents (baseline window); the shared $\pm(0.02\text{--}0.04)\ \Omega$ structure sets the systematic scale. (c) Synthetic validation of the drift decomposition: recovered δT distributions without (blue) and with (orange) a 20% intrinsic polarity asymmetry; dashed line marks the injected 0.3 K.

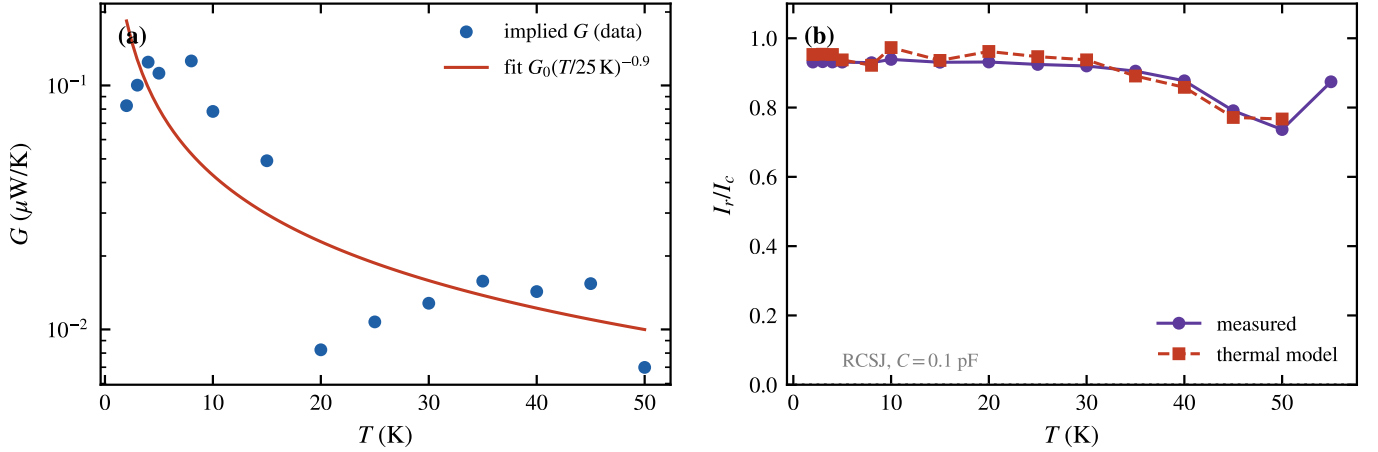


FIG. 6. Thermal power-balance analysis. (a) Effective thermal conductance implied by the measured retrapping points; the line is the smooth interpolation used for the forward prediction. Points for $T \geq 20$ K are lower bounds (see text). (b) Measured $I_r/I_c(T)$ versus the parameter-free forward prediction of the calibrated model; the dotted line is the RCSJ expectation for $C = 0.1$ pF.

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